CS701 – Software Architectures

Unit – I

- **Main**
  - Software Development Methodologies
  - Software Architecture
  - Architectural Patterns
- **Detail**
  - Software quality models – Development models – Issues
  - Introduction – Evolution of software architecture
  - Software components – Connectors
  - Architecture frameworks
  - Architecture business cycle
  - Architectural patterns – Reference model

Unit – II

- **Main**
  - Architecture Models
  - Architectural Styles
  - Modern Architectures
- **Detail**
  - Structural – Framework – Dynamic – Process models
  - Dataflow – Pipes & filters
  - Call & return – Data centered – Layered architecture
  - Agent based architecture
  - Micro-services – Reactive – REST architecture

Unit – III

- **Main**
  - Architecture Implementation
  - Middleware
  - UML
- **Detail**
  - ADLs
  - Struts – Hibernate – Node JS – Angular JS
  - J2EE (JSP – Servlets – EJBs)
  - JDBC – JNDI – JMS – RMI – CORBA
  - Role of UML in software architecture

Unit – IV

- **Main**

    - Architecture Analysis
    - Architecture Design
    - Reuse
- **Detail**
    - Requirements for architecture
    - Architecture life-cycle
    - CBAM – ATAM
    - ARID – ADD
    - Architecture reuse
    - Domain specific software architecture

Unit – V

- **Main**
    - Architecture Documentation
    - Documentation Principles
    - Documentation Templates
- **Detail**
    - Principles of sound documentation
    - Refinement – Context diagrams – Variability
    - Software interfaces
    - Behavior documentation
    - Seven-part documentation template

CS702 (A) – Computational Intelligence

Unit – I

- **Main**
  - Computational Intelligence
  - Learning Models
  - Neural Networks
- **Detail**
  - Types – Components of CI
  - Learning / Training models
  - Parametric – Nonparametric models
  - Feed forward – Feedback networks

Unit – II

- **Main**
  - Fuzzy Systems
  - Fuzzy Logic
  - Fuzzy Control
- **Detail**
  - Fuzzy sets – Operations – Membership functions
  - Fuzzy relations – Measures
  - Fuzzy rules – Inferencing
  - Fuzzyfication – Defuzzyfication

Unit – III

- **Main**
  - Genetic Algorithms
  - Genetic Operators
  - Genetic Modeling
- **Detail**
  - Basic genetics – Working principle
  - Encoding – Fitness function
  - Selection – Crossover – Mutation
  - Benefits of GA

Unit – IV

- **Main**
  - Rough Set Theory
  - HMM
  - Decision Trees
- **Detail**

    - Set approximation – Rough membership
    - Attributes – Optimization
    - Hidden Markov Models
    - Decision tree model

Unit – V

- **Main**
    - Swarm Intelligence
    - Swarm Techniques
    - Applications
- **Detail**
    - Ant colony optimization
    - Particle swarm optimization
    - Bee colony optimization
    - Applications of CI

CS702 (B) – Deep & Reinforcement Learning

Unit – I

- **Main**
  - Deep Learning Basics
  - Optimization Algorithms
  - RNN Models
- **Detail**
  - Activation functions – Gradient descent variants
  - AdaGrad – RMSProp – Adam
  - RNN – BPTT – GRU – LSTM
  - Attention mechanisms

Unit – II

- **Main**
  - Autoencoders
  - Regularization
  - Normalization
- **Detail**
  - Sparse – Denoising – Contractive autoencoders
  - Bias variance tradeoff – Early stopping
  - Dropout – Batch – Instance – Group normalization

Unit – III

- **Main**
  - CNN Architectures
  - Visualization
  - Trends
- **Detail**
  - LeNet – AlexNet – VGG – GoogLeNet – ResNet
  - Guided backpropagation – Deep dream – Deep art
  - Recent trends

Unit – IV

- **Main**
  - Reinforcement Learning
  - MDP
  - Value Methods
- **Detail**
  - Bandit algorithms – Policy gradient
  - Bellman optimality

    - Value iteration – Policy iteration
    - Q-learning – TD learning

Unit – V

- **Main**
    - Advanced RL
    - Actor-Critic
    - Imitation Learning
- **Detail**
    - Deep Q learning – DQN
    - Hierarchical RL – POMDPs
    - Inverse RL – GAN based imitation

CS702 (C) – Wireless & Mobile Computing

Unit – I

- **Main**
  - Traditional Networks
  - TCP/IP
  - VLAN & IPv6
- **Detail**
  - LAN – MAN – WAN – Internet
  - ARP – IP – ICMP – DNS
  - NAT – VLANs
  - IPv6 addressing

Unit – II

- **Main**
  - Routing Protocols
  - TCP
  - UDP
- **Detail**
  - BGP – RIP – OSPF
  - TCP segment – Flow control – Congestion
  - TCP variants – Tahoe – Reno – Sack
  - UDP format

Unit – III

- **Main**
  - Wireless LAN
  - Mobile IP
  - Ad hoc Routing
- **Detail**
  - IEEE 802.11 – MAC issues
  - Spread spectrum – Security
  - Mobile IP registration – Tunneling
  - AODV – DSDV – DSR – ZRP

Unit – IV

- **Main**
  - Mobile Transport
  - Cellular Networks
  - GSM
- **Detail**

    - I-TCP – M-TCP – S-TCP
    - Cellular vs WLAN
    - GSM architecture – Services – Roaming

Unit – V

- **Main**
    - Mobile OS
    - SDKs
    - M-Commerce
- **Detail**
    - iOS – Android OS
    - Mobile SDKs
    - Mobile payment – Security issues

CS702 (D) – Big Data

Unit – I

- **Main**
  - Big Data Concepts
  - Characteristics
  - Infrastructure
- **Detail**
  - Types – Evolution – Challenges
  - Traditional vs Big Data
  - Analytics – Desired properties

Unit – II

- **Main**
  - Hadoop
  - HDFS
  - MapReduce
- **Detail**
  - Hadoop ecosystem – YARN
  - RDBMS vs Hadoop
  - Resource management

Unit – III

- **Main**
  - Hive
  - Pig
  - ETL
- **Detail**
  - Hive architecture – HQL
  - Pig anatomy – Execution model
  - Operators – Functions

Unit – IV

- **Main**
  - NoSQL
  - Architecture Patterns
  - MongoDB
- **Detail**
  - NoSQL drivers
  - Big data management
  - MongoDB introduction

Unit – V

- **Main**
  - Social Network Mining
  - Graphs
  - Recommender Systems
- **Detail**
  - Community detection
  - Graph clustering
  - Recommendation basics

CS703 (A) – Cryptography & Information Security

Unit – I

- **Main**
  - Mathematical Background
  - Classical Cryptography
  - Symmetric Encryption
- **Detail**
  - Number theory – Modular inverse – Euler theorem
  - Substitution cipher – Playfair
  - DES – Triple DES – Modes – Stream cipher

Unit – II

- **Main**
  - Public Key Cryptography
  - RSA
  - Elliptic Curve
- **Detail**
  - Diffie-Hellman – Discrete log
  - RSA signatures – Schnorr
  - Chinese remainder theorem

Unit – III

- **Main**
  - Authentication
  - Hash Functions
  - Secure Channels
- **Detail**
  - MAC – Digital signatures
  - SHA – DSS
  - Kerberos

Unit – IV

- **Main**
  - Network Security
  - Email Security
  - Web Security
- **Detail**
  - Firewalls – IDS

    - PGP – S/MIME
    - IPSec – SSL/TLS – SET

Unit – V

- **Main**
    - Security Tools
    - Malware Detection
    - DoS Tools
- **Detail**
    - Scanners – Spoofing – Steganography
    - Trojan detection
    - DoS attack tools

CS703 (B) – Data Mining & Warehousing

Unit – I

- **Main**
  - Data Warehousing
  - Preprocessing
  - Design
- **Detail**
  - Architecture – ETL – Metadata
  - Cleaning – Integration – Reduction
  - Multidimensional models

Unit – II

- **Main**
  - OLAP
  - Hardware Design
  - Security
- **Detail**
  - OLAP operations – Servers
  - Backup – Recovery

Unit – III

- **Main**
  - Data Mining
  - Similarity Measures
  - Fuzzy Logic
- **Detail**
  - Data types – Distributions
  - Mining vs KDD
  - Fuzzy sets

Unit – IV

- **Main**
  - Supervised Learning
  - Classification
  - Neural Networks
- **Detail**
  - Decision trees – Rule based
  - Probabilistic classifiers

Unit – V

- **Main**
  - Clustering
  - Association Rules
  - Algorithms
- **Detail**
  - BIRCH – DBSCAN – CURE
  - Apriori – FP Growth

CS703 (C) – Agile Software Development

Unit – I

- **Main**
  - Agile Fundamentals
  - Agile Models
  - Principles
- **Detail**
  - Agile manifesto
  - Scrum – XP – Kanban – Lean

Unit – II

- **Main**
  - Agile Planning
  - Release Planning
  - Monitoring
- **Detail**
  - User stories – Backlogs
  - Estimation – Retrospectives

Unit – III

- **Main**
  - Scrum Framework
  - Scrum Artifacts
  - Case Study
- **Detail**
  - Sprint planning – Review
  - Burn down charts

Unit – IV

- **Main**
  - Extreme Programming
  - XP Practices
  - Adoption
- **Detail**
  - Pair programming
  - Refactoring – Velocity

Unit – V

- **Main**

    - Agile Design
    - QA
    - TDD
- **Detail**
    - Continuous integration
    - Automated testing

CS703 (D) – Disaster Management

Unit – I

- **Main**
  - Disasters
  - Types
  - Impacts
- **Detail**
  - Earthquake – Flood – Fire – Drought
  - Social – Economic – Environmental impacts

Unit – II

- **Main**
  - Disaster Risk Reduction
  - Institutions
  - Early Warning
- **Detail**
  - Disaster cycle
  - SDMA – NDMA

Unit – III

- **Main**
  - Disasters & Development
  - Climate Change
  - Indigenous Knowledge
- **Detail**
  - Vulnerability – Adaptation

Unit – IV

- **Main**
  - Disaster Management in India
  - Relief Components
  - GIS
- **Detail**
  - DM Act – Policies
  - Damage assessment

Unit – V

- **Main**
  - Case Studies

    - Field Work
    - Applications
- **Detail**
    - Earthquake – Flood – Fire – Drought cases